

Each of these facts follows immediately by noting that an element that belongs to the first set in each pair of sets also belongs to the second set in that pair. $\square$

Example 9: The set of integers with squares less than 100 is not a subset of the set of nonnegative integers because -1 is in the former set [as $(-1)^2 < 100$] but not the latter set. The set of people who have taken discrete mathematics at your school is not a subset of the set of all computer science majors at your school if there is at least one student who has taken discrete mathematics who is not a computer science major. $\square$

Theorem 1: For every set S , (i) $\emptyset \subseteq S$ and (ii) $S \subseteq S$

Proof: Let c be an arbitrary element. By defn. of $\emptyset$, $c \in \emptyset$ is False. By defn. of condition, $c \in \emptyset \rightarrow c \in S$ is True.

By universal generalization, $\forall x (x \in \emptyset \rightarrow x \in S)$ is True.

$\therefore$ By Definition 3, $\emptyset \subseteq S$ (i)

Let x be an arbitrary element, where $x \in S$. By defn. of conditional, $x \in S \rightarrow x \in S$ is True.

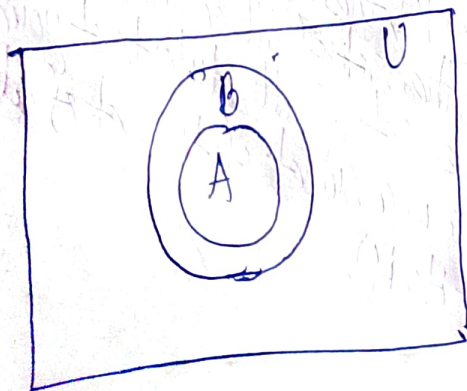
By universal generalization, $\forall x (x \in S \rightarrow x \in S)$ is True.

$\therefore$ By Definition 3, $S \subseteq S$ (ii) $\square$

$A \subset B$: A is a proper subset of B .

For $A \subset B$ to be true, it must be the case that $A \subseteq B$ and there must exist an element x of B that is not an element of A .

$A \subseteq B$ if and only if $\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$ is true.



To show that two sets A and B are equal, show that $A \subseteq B$ and $B \subseteq A$.